

Institute of Space Technology	Form No	IST-Dean-F-23/00
FYP Selection Form	Issue Date	
Grading Policy on Final Year Projects	Department	Dean Office

Final Year Project (FYP) Defense Proposal Form

University/Institute Detail:

Name of University/Institution	Institute of Space Technology	Physical Address:	1, Islamabad Highway, Islamabad. P.O 44000
Telephone & Fax No:	+92-51-9075100 +92-51-9273310	Department Name:	Space Science
City:	Islamabad	Province:	Capital Territory (ICT)

Nominated Project Details:-

Project Supervisor Name and Designation:	Dr. Munawar Shah Assistant Professor	Contact Details:	Munawar.shah@ist.edu.pk
Project Supervisor Qualification:	Ph.D. in Geodesy and GNSS	No of Publications of Supervisors:	over 100 peer-reviewed papers
Students Name(s):	Mashaf Majeed Janjua Haider Taqveen	Students Mobile No:	+923205409300 +923259514052
Students CGPA	2.31 2.65	Students Email:	230601017@ist.edu.pk 230601020@ist.edu.pk
Degree Program/ Title:	Bs Space Science	Area of Specialization:	RS/GIS

Final Year Project Details:

A. Project Title:	A Multi-Agent Deep Reinforcement Learning Approach in a Geospatial Digital Twin
B. Project Start Date:	
C. Project Finish Date:	
D. Project Summary (less than 200 words)	

The use of Unmanned Aerial Vehicles (UAVs) is growing in areas such as search and rescue, disaster assessment, environmental monitoring, mapping, and surveillance. As the number of

Institute of Space Technology	Form No	IST-Dean-F-23/00
FYP Selection Form	Issue Date	
Grading Policy on Final Year Projects	Department	Dean Office

UAVs increases, coordinating multiple UAVs to perform tasks efficiently becomes more complex. Testing swarm-based systems with physical drones can also be costly, time-consuming, and difficult to conduct repeatedly in controlled conditions. This project proposes an AI-enabled Digital Twin for simulating UAV swarms, allowing their autonomous behavior and coordination to be trained, tested, and evaluated in different dynamic environments.

The proposed Digital Twin will represent a UAV operating area containing multiple drone agents, obstacles, mission targets, and environmental constraints. Reinforcement learning and multi-agent learning techniques will be investigated to allow the UAVs to learn cooperative behaviors, including navigation, collision avoidance, area coverage, formation maintenance, and target reaching. Different operating conditions will be introduced during simulation, such as changing obstacles, UAV failures, communication limitations, and changes in mission requirements. These scenarios will help assess how the swarm responds to changes in the environment and mission requirements.

The swarm's performance will be evaluated using parameters such as mission completion time, area coverage, collision rate, energy consumption, and adaptability. GIS and geographical data will also be incorporated into the virtual environment to create more realistic operating conditions.

E. Project Objectives:

1. To develop a digital-twin-based simulation environment for autonomous multi-UAV swarm operations.
2. To develop and simulate multiple autonomous UAV agents capable of cooperative navigation and mission execution.
3. To investigate reinforcement learning and multi-agent learning techniques for UAV swarm coordination.
4. To simulate changing environmental and mission conditions, including obstacles, UAV failures, communication limitations, and changing targets.
5. To evaluate swarm performance using mission completion time, area coverage, collision rate, energy consumption, and adaptability.
6. To develop a visualization and monitoring interface for observing the digital twin, UAV swarm behavior, mission state, and performance results.

Institute of Space Technology	Form No	IST-Dean-F-23/00
FYP Selection Form	Issue Date	
Grading Policy on Final Year Projects	Department	Dean Office

F. Project Implementation Method			
The project will be developed as a software-based digital-twin and simulation framework. The first stage will involve creating a virtual operating environment containing the main features required for UAV missions, such as terrain, obstacles, restricted areas, and mission targets. Where suitable data are available, GIS and geographical information will be incorporated to make the simulated environment more representative of real operating areas. A multi-UAV simulation module will represent the individual drones and their states. The simulation will maintain information such as UAV position, velocity, altitude, energy state, mission status, and communication status. The Digital Twin will maintain a virtual representation of the environment and the UAV swarm and update these states throughout the simulation. An AI-based control framework will then be investigated using reinforcement learning and multi-agent reinforcement learning. Each UAV agent will receive observations from the simulated environment and select actions according to its current state and mission requirements. The learning process will focus on improving mission performance while reducing collisions and unnecessary energy consumption. The system will include different scenarios for testing swarm behavior under changing conditions. These will include moving or changing obstacles, UAV failures, communication limitations, and changes in mission objectives. After training, the AI-based swarm will be evaluated using predefined performance measures and compared with selected baseline control strategies. A visualization interface will be developed as part of the final system. It will allow the user to observe the virtual environment, UAV states, mission progress, swarm behavior, and performance results during and after simulation.			
G. Key Milestones of the Project with Dates			
S.No.	Elapsed time since start of the project	Milestone	Deliverable
1.	Month 1	Literature Review & System Design	Literature review, requirements and system architecture
2.	Month 2-3	Digital Twin & UAV Simulation Development	Functional virtual environment and multi-UAV simulator
3.	Month 4-5	AI/Swarm Training	Trained AI-based swarm-control model

Institute of Space Technology	Form No	IST-Dean-F-23/00
FYP Selection Form	Issue Date	
Grading Policy on Final Year Projects	Department	Dean Office

4.	Month 6-7	Dynamic Scenario Testing & Evaluation	Experimental results and performance analysis
5.	Month 8	Integration & Final Demonstration	Final digital twin, visualization interface, report and demonstration

--

H. Final Deliverable of the Project:	Software System Software Simulation results Comparative Study Simulator Design
---	---

I. Please Specify Technical Details of Final Deliverable	
---	--

The final deliverable will be a software-based AI-enabled Digital Twin platform for simulation and evaluation of autonomous multi-UAV swarm operations.

The system will include the following components:

1. Virtual Environment: Represents the UAV operating area, including terrain, obstacles, restricted regions, and mission targets.
2. Digital Twin Module : Maintains the virtual state of the environment and the UAV swarm during simulation.
3. Multi-UAV Simulation Module: Simulates multiple autonomous UAV agents and their movement and operational states.
4. AI Swarm-Control Module: Uses reinforcement learning and multi-agent reinforcement learning to coordinate autonomous UAVs.
5. Dynamic Scenario Module: Simulates changing obstacles, UAV failures, communication constraints, and mission changes to test swarm adaptability.
6. Mission Management Module: Defines mission objectives and manages their execution by the UAV swarm.
7. Performance Evaluation Module: Measures mission completion time, area coverage, collision rate, energy consumption, and adaptability.
8. Visualization Interface: Displays the UAV positions, environmental state, swarm behavior, mission progress, and performance results.
9. Experimental Evaluation: Provides results comparing the AI-based swarm behavior with selected baseline approaches.

J. Equipment required for making prototype/working model:	
--	--

No specialized physical hardware is required for the core implementation. The project will primarily use a computer or laptop capable of running the simulation, AI training, GIS processing, and visualization software.

The required resources include:

1. Development workstation or laptop

Institute of Space Technology	Form No	IST-Dean-F-23/00
FYP Selection Form	Issue Date	
Grading Policy on Final Year Projects	Department	Dean Office

2. GPU capability for AI model training, where required
3. Internet connectivity for obtaining datasets and software packages
4. GIS and geospatial datasets
5. UAV simulation and AI software frameworks A physical UAV or hardware-in-the-loop equipment may be used as an optional extension or demonstration if suitable hardware is available.

K. Benefits of the Project

The proposed system will provide a controlled and repeatable environment for developing and evaluating autonomous UAV swarm algorithms without requiring multiple physical drones.

The expected benefits include:

1. Lower cost and reduced risk compared with testing a swarm of physical UAVs.
2. Repeatable experiments under controlled environmental and mission conditions.
3. Evaluation of swarm behavior under changing and uncertain conditions.
4. Investigation of AI-based cooperative navigation and decision-making.
5. Quantitative evaluation of different swarm-control approaches.
6. Potential use in applications such as search and rescue, disaster assessment, environmental monitoring, mapping, surveillance, and infrastructure inspection.
7. Use of GIS and geographical information to create more realistic simulation environments.
8. A foundation for future hardware-in-the-loop testing and experiments with physical UAVs.

It is certified that the FYP titled “A Multi-Agent Deep Reinforcement Learning Approach in a Geospatial Digital Twin” has been approved and is being undertaken by the above-mentioned students as their Final Year Project.

It is undertaken that the undersigned have understood the “terms & conditions” of the program, and further reiterate that, if the subject FYP is approved for funding, the disbursed funds shall be utilized as per “terms & conditions” and the undersigned will be liable to reimburse/refund the unutilized amount and other cost not approved by the IST, if any.

It is further undertaken that after utilization of funds, the said “Fund Utilization Report” shall be furnished along with the other required deliverables as and when required by the IST.

1. Name, Designation& Signature of Supervisor: _____

2. Name & Signatures of HoD: _____

Institute of Space Technology	Form No	IST-Dean-F-23/00
FYP Selection Form	Issue Date	
Grading Policy on Final Year Projects	Department	Dean Office